SIMPLE RAG PROJECT

60-topic requirements analysis and workflow guide

What was analyzed	The pasted documentation prompt
Number of topics	60 distinct requirements
Page structure	One topic per page after this cover
Scope note	Repository-specific values must be verified from source code.

Reading rule: this is a requirements breakdown, not a claim that the unseen repository already implements every item. Use the required labels NOT IMPLEMENTED IN THIS PROJECT and PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED when evidence is missing.

TOPIC 1 OF 60

Project purpose

What this requirement means

Define what the RAG project is expected to do and what problem it solves.

Workflow

Goal -> user question -> grounded answer

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 2 OF 60

Repository inspection

What this requirement means

Inspect the complete repository before writing implementation documentation.

Workflow

List files -> read source/config -> trace execution -> record evidence

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 3 OF 60

Frontend identification

What this requirement means

Find the user-facing application and explain how it collects and displays questions and answers.

Workflow

User -> UI input -> request -> response -> UI output

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 4 OF 60

Backend identification

What this requirement means

Find the server-side application that handles ingestion, retrieval, and generation.

Workflow

HTTP request -> route -> service -> response

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 5 OF 60

Languages and frameworks

What this requirement means

Inventory programming languages, frameworks, and major libraries.

Workflow

Manifest files -> dependencies -> runtime components

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 6 OF 60

LLM identification

What this requirement means

Identify the exact language model used to generate answers.

Workflow

Prompt + context -> model -> answer

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 7 OF 60

Embedding model identification

What this requirement means

Identify the model that converts text and queries into vectors.

Workflow

Text/query -> embedding model -> vector

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 8 OF 60

Vector database identification

What this requirement means

Identify the vector store and explain its role in retrieval.

Workflow

Chunk + vector + metadata -> index -> search

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 9 OF 60

Normal database identification

What this requirement means

Check whether a conventional database is used for users, metadata, jobs, or application state.

Workflow

Application event -> database read/write -> result

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess. If absent, state: NOT IMPLEMENTED IN THIS PROJECT. If proposed for production, state: PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED.

TOPIC 10 OF 60

Document source inventory

What this requirement means

List exactly where knowledge comes from: files, URLs, records, uploads, or other sources.

Workflow

Source -> loader -> extracted content -> indexing

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 11 OF 60

Document loaders

What this requirement means

Explain the loader used for each supported document type.

Workflow

File/source -> loader -> document objects

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 12 OF 60

Text extraction

What this requirement means

Show how a raw document becomes usable text.

Workflow

Document -> pages/records -> text -> metadata

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 13 OF 60

Metadata preservation

What this requirement means

Explain which metadata travels with each chunk and why it matters.

Workflow

Source metadata -> chunk metadata -> filters/citations/debugging

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 14 OF 60

Cleaning and normalization

What this requirement means

Identify whitespace cleanup, encoding fixes, or text normalization.

Workflow

Raw text -> cleaned text -> chunker

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 15 OF 60

Chunk concept

What this requirement means

Explain a chunk as a small searchable piece of a document.

Workflow

Long document -> smaller passages -> searchable units

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 16 OF 60

Chunk size

What this requirement means

Record the configured maximum chunk size and its unit.

Workflow

Text -> sized chunks -> embeddings

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 17 OF 60

Chunk overlap

What this requirement means

Record how much neighboring chunks repeat.

Workflow

Chunk N overlaps with chunk N+1 -> continuity

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 18 OF 60

Chunk quality trade-offs

What this requirement means

Explain how overly large or tiny chunks affect retrieval and generation.

Workflow

Chunking -> retrieval precision/context -> answer quality

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 19 OF 60

Embedding fundamentals

What this requirement means

Explain why text is converted into numbers for semantic search.

Workflow

Text meaning -> vector representation -> nearest neighbors

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 20 OF 60

Embedding input/output

What this requirement means

Document the exact input text and vector output path.

Workflow

Text -> embedding call -> vector -> storage/search

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 21 OF 60

Embedding dimensions and cost

What this requirement means

Report vector dimension and operational cost only when confirmed.

Workflow

Model choice -> dimension/cost/latency -> database settings

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 22 OF 60

Vector storage schema

What this requirement means

Describe how each indexed item is represented.

Workflow

ID + text + vector + metadata -> vector store

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 23 OF 60

Collection/index lifecycle

What this requirement means

Explain creation, loading, reuse, update, and deletion behavior.

Workflow

Startup/ingest -> create/open -> write -> query

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 24 OF 60

Persistence

What this requirement means

Explain whether vectors survive process restarts.

Workflow

Store -> restart -> data available or missing

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 25 OF 60

Similarity search

What this requirement means

Explain how a query vector is compared with stored vectors.

Workflow

Question -> query vector -> distance -> ranked chunks

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 26 OF 60

Top-K retrieval

What this requirement means

Explain K as the number of candidate chunks returned.

Workflow

Ranked results -> first K chunks -> context

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 27 OF 60

Retriever role

What this requirement means

Distinguish the vector database from the retriever abstraction.

Workflow

Retriever settings -> vector store query -> documents

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 28 OF 60

Retriever filters

What this requirement means

Check for metadata filters, namespaces, tenant filters, or source restrictions.

Workflow

Query + filters -> narrowed search -> results

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 29 OF 60

Similarity thresholds

What this requirement means

Check whether weak matches are rejected before prompting the LLM.

Workflow

Results -> score threshold -> accepted context

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 30 OF 60

Hybrid search

What this requirement means

Check whether keyword and vector search are combined.

Workflow

Query -> lexical + vector search -> merged ranking

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess. If absent, state: NOT IMPLEMENTED IN THIS PROJECT. If proposed for production, state: PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED.

TOPIC 31 OF 60

Reranking

What this requirement means

Explain the optional second ranking stage.

Workflow

Query -> initial top 20 -> reranker -> best 5 -> LLM

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess. If absent, state: NOT IMPLEMENTED IN THIS PROJECT. If proposed for production, state: PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED.

TOPIC 32 OF 60

Query intake

What this requirement means

Trace how a user question enters the system.

Workflow

Request body -> validation -> query variable

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 33 OF 60

Query embedding

What this requirement means

Explain how the question is embedded for retrieval.

Workflow

Question -> embedding model -> query vector

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 34 OF 60

Context assembly

What this requirement means

Explain how retrieved chunks are combined for the prompt.

Workflow

Retrieved chunks -> ordered context -> prompt input

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 35 OF 60

Prompt template

What this requirement means

Document the exact instruction template used around the context.

Workflow

Question + context + rules -> formatted prompt

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 36 OF 60

Prompt safety

What this requirement means

Check for instructions that reduce prompt injection or unsupported claims.

Workflow

Untrusted text -> boundaries/instructions -> safer generation

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 37 OF 60

LLM call

What this requirement means

Explain the code path that sends the prompt to the model.

Workflow

Formatted prompt -> client call -> model output

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 38 OF 60

Answer generation

What this requirement means

Explain how the model turns context into a response.

Workflow

Grounded prompt -> generation -> answer

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 39 OF 60

RAG chain

What this requirement means

Describe how retrieval, prompt construction, and generation connect.

Workflow

Question -> retrieve -> format -> generate -> return

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 40 OF 60

Response format

What this requirement means

Document the API or UI response structure.

Workflow

Answer + metadata/errors -> JSON/UI

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 41 OF 60

API endpoints

What this requirement means

Inventory routes used for upload, indexing, querying, health, or administration.

Workflow

Client -> route -> handler -> service -> response

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 42 OF 60

Authentication

What this requirement means

Check whether the project protects APIs or documents.

Workflow

Identity/token -> authorization -> operation

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess. If absent, state: NOT IMPLEMENTED IN THIS PROJECT. If proposed for production, state: PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED.

TOPIC 43 OF 60

Environment variables

What this requirement means

List required secrets, model names, URLs, paths, and tuning values.

Workflow

Environment/config -> startup -> runtime behavior

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 44 OF 60

Configuration files

What this requirement means

Explain where settings live and which values override others.

Workflow

Config + environment -> resolved settings

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 45 OF 60

Docker configuration

What this requirement means

Check for Dockerfiles, compose files, volumes, ports, and services.

Workflow

Image build -> container startup -> network/storage

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 46 OF 60

Deployment configuration

What this requirement means

Check for hosting, process, reverse proxy, or cloud deployment instructions.

Workflow

Code -> build -> deploy -> service

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 47 OF 60

Error handling

What this requirement means

Trace failures in loading, embedding, retrieval, and generation.

Workflow

Failure -> exception handling -> log/status -> user response

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 48 OF 60

Logging

What this requirement means

Identify what events and diagnostics are recorded.

Workflow

Runtime event -> logger -> output

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 49 OF 60

Testing strategy

What this requirement means

Inventory unit, integration, API, retrieval, and end-to-end tests.

Workflow

Test input -> component/system -> expected result

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess. If absent, state: NOT IMPLEMENTED IN THIS PROJECT. If proposed for production, state: PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED.

TOPIC 50 OF 60

Ingestion workflow

What this requirement means

Summarize the offline or startup indexing process.

Workflow

Documents -> load -> extract -> clean -> chunk -> embed -> store

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 51 OF 60

Question-answer workflow

What this requirement means

Summarize the online query path.

Workflow

Question -> embed -> retrieve -> prompt -> LLM -> response

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 52 OF 60

Normal LLM vs RAG

What this requirement means

Compare direct generation with context-grounded generation.

Workflow

Direct: user -> LLM; RAG: user -> retrieve -> context -> LLM

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 53 OF 60

Hallucination and grounding

What this requirement means

Explain how unsupported answers can occur and how evidence helps.

Workflow

Weak evidence -> model guess; strong evidence -> grounded answer

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 54 OF 60

Private and changing data

What this requirement means

Explain why RAG is useful for internal or frequently updated knowledge.

Workflow

Documents -> re-index -> retrieve current evidence -> answer

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 55 OF 60

Architecture diagram

What this requirement means

Create a diagram that matches actual components and data flow.

Workflow

User -> frontend -> API -> retrieval/generation -> response

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

Beginner explanation

What this requirement means

Explain the project without assuming knowledge of embeddings or vector stores.

Workflow

Plain concept -> example -> workflow

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

TOPIC 57 OF 60

Technical explanation

What this requirement means

Explain the same system with precise engineering vocabulary.

Workflow

Components -> interfaces -> data structures -> runtime sequence

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

Interview explanation

What this requirement means

Prepare a concise 1-2 minute explanation of the project.

Workflow

Problem -> architecture -> retrieval -> generation -> trade-offs

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.

Evidence and gaps

What this requirement means

Separate confirmed behavior, unknowns, and recommendations.

Workflow

Source evidence -> implemented / not implemented / recommended

Documentation guidance

Separate confirmed behavior, unknowns, and recommendations. The prompt explicitly requires this distinction.

TOPIC 60 OF 60

Production readiness review

What this requirement means

Turn the requirements into a final gap analysis.

Workflow

Evidence -> risks -> prioritized improvements

Documentation guidance

Trace this behavior to source files, functions, configuration, dependencies, and tests. Do not guess.